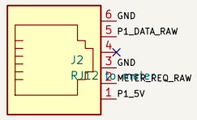
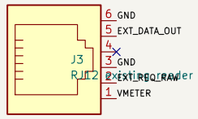
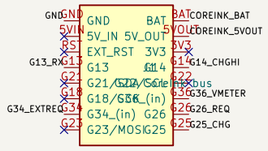
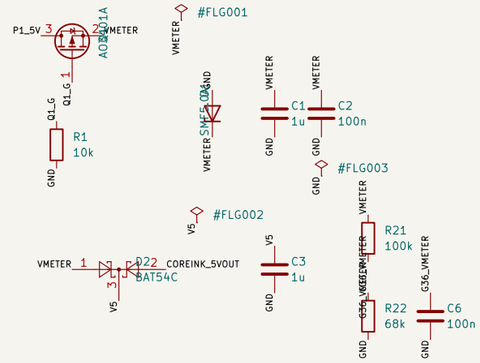
## 1 Connectors



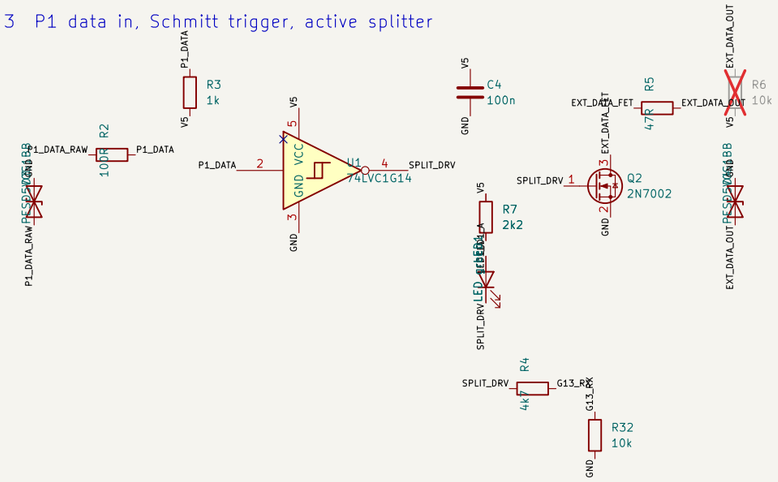



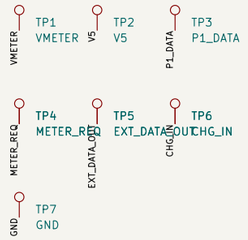

## 2 Meter 5 V, protection, V5



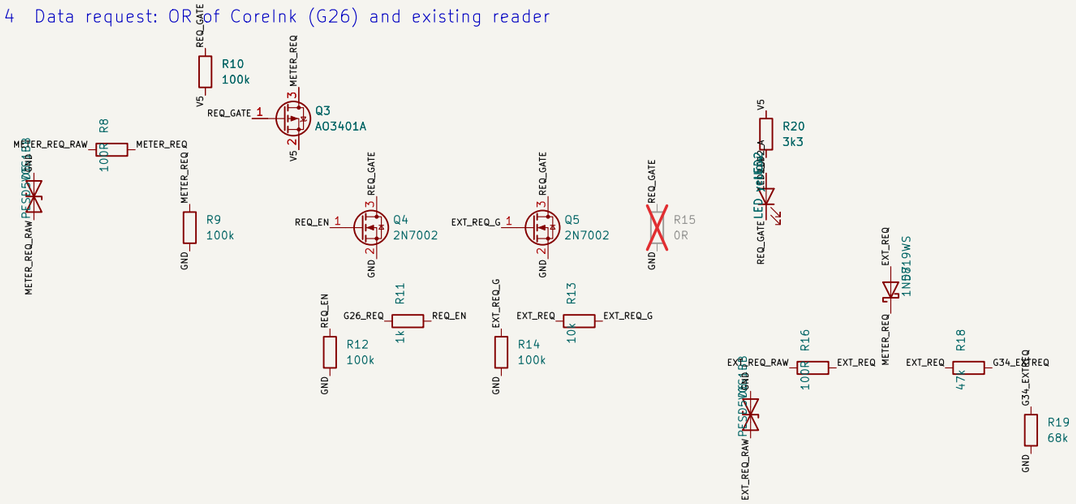

## 3 P1 data in, Schmitt trigger, active splitter



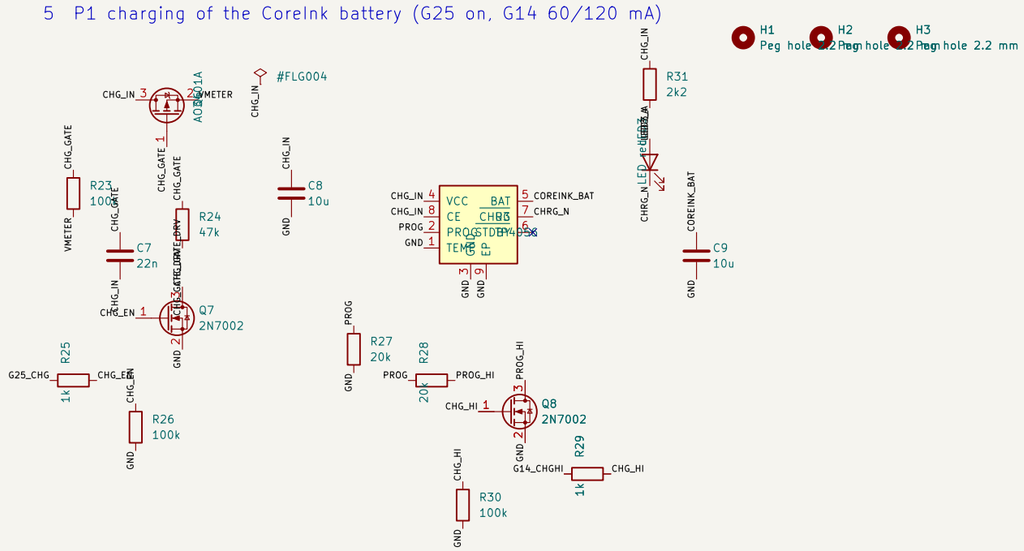

## 6 Test points




## 4 Data request: OR of CoreInk (G26) and existing reader




## 5 P1 charging of the CoreInk battery (G25 on, G14 60/120 mA)




## 7 Enclosure peg holes

H1 Peg hole 2.2 mm
H2 Peg hole 2.2 mm
H3 Peg hole 2.2 mm

NeedForHeat P1-BASE v3.0 – design rationale in docs/design-v3.md

Firmware interface: G13 = P1 RX (not inverted); G26 = data request (high = request);

G25 = P1 charging on (high); G14 = charge current 120 mA (high) / 60 mA (low);

G36 = meter 5 V via 100k/68k (x0.405); G34 = existing reader requests (47k/68k).

DNP: R6 = pull-up for readers without own pull-up; R15 = always request data.

Continues the P1-BASE by Research group Energy Transition, Windesheim
Licensed under CERN-OHL-P v2
M5Stack CoreInk extension: P1 smart meter reader with active splitter
**energiegedrag.nl**

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**Title: NeedForHeat P1-BASE**

| Size: A3 | Date: 2026-11-13 | **Rev: 3.0** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 1/1 |